

SLC: Securing the Wallet & Unlocking Growth

for real-time proof of possession



The Reality of Wallet Security Today



Account Takeover (ATO) is the primary threat

- Direct access to accounts and funds
- High-impact, high-frequency attacks



Traditional authentication is breaking down

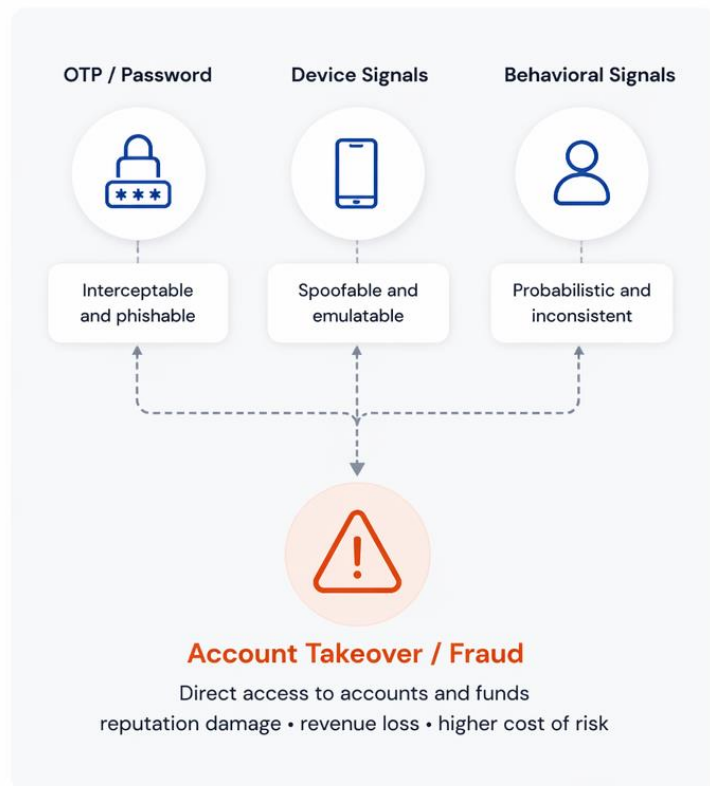
- OTPs are easily intercepted or socially engineered
- Device and behavioral signals can be spoofed



AI is accelerating attack sophistication

- Automated phishing and impersonation at scale
- Faster, more adaptive attack patterns

Wallet platforms are securing high-value transactions with methods that were never designed for this level of risk.



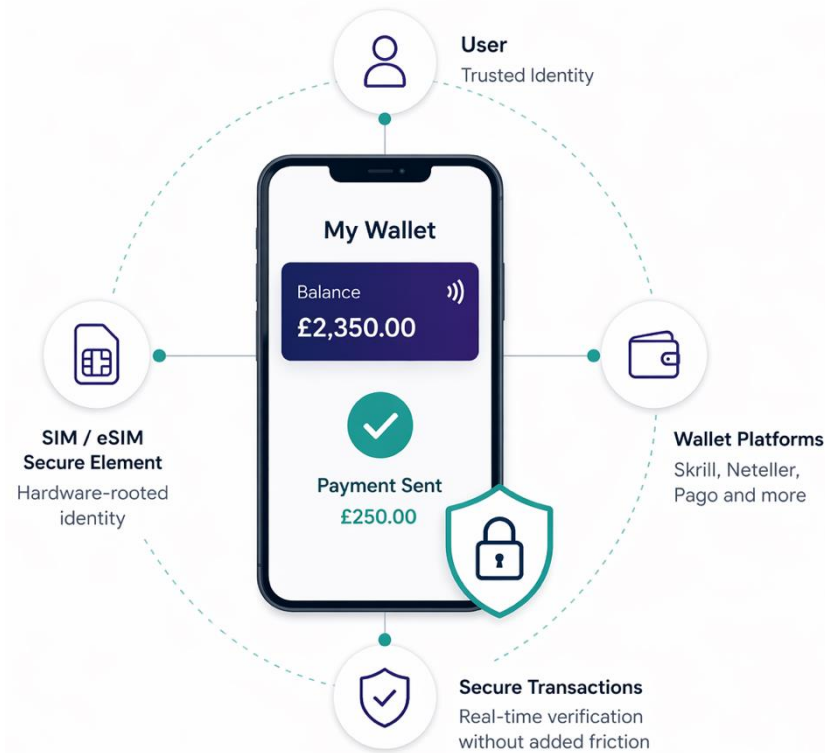
SLC- A New Trust Layer for Digital Wallets

A hardware-rooted identity layer for **real-time proof of possession**

SLC is building a hardware-rooted identity layer, anchoring authentication directly to the SIM/eSIM secure element.

This enables deterministic proof of device possession— independent of the OS or application layer.

For wallet platforms **where identity = access to funds**— this introduces a fundamentally stronger model for authentication.



- Eliminates reliance on probabilistic signals (OTP, device, behavior)
- Reduces fraud exposure in account access and transactions
- Enables secure authentication without added friction

From Probabilistic Authentication to Deterministic Trust

SLC fundamentally changes how digital wallets verify identity.



Hardware-Rooted Identity

- Authentication anchored to the SIM/eSIM secure element
- Not credentials or device signals



Deterministic Verification

- Every transaction is cryptographically verified
- Confirms the real device is present in real time



Real-Time Trust Signals

- Continuous network & device validation
- Non-replayable proof of trust



Identity is no longer inferred—
it is cryptographically verified at the hardware level.



Identity anchored in the SIM Secure Element— independent of the OS and application layer

Real-time proof of device possession

A private key generated inside the Secure Element at onboarding.

- Never transmitted
- Never exported
- Signs every transaction in real time



Deterministic proof of possession and presence—every time.

WHERE THE THREAT LIVES

APPLICATION LAYER

AI-accessible • malleable

OPERATING SYSTEM

AI-accessible • exploitable

SLC GUARDIAN

SIM Secure Element



OS-independent • secured



The Hidden Power of the **SIM Layer**

SLC operates outside the OS trust boundary—within the SIM Secure Element

LAYER 1 — Application / OS

Passkeys · OTP · Push notifications · Behavioral biometrics · Device attestation (varies by OEM)

Reachable by malware, AI attacks, OS compromise, jailbreak

LAYER 2 — Device Secure Hardware

TEE / Android Keystore (mandatory) · StrongBox / Titan M2 (Pixel only, recommended for others)

Device-dependent, inside device trust boundary, OEM variation across Android ecosystem

LAYER 3 — SIM / eSIM Secure Execution Environment

Hardware-bound key pair · IMEI/IMSI binding · OTA channel · Independent of device OEM · Consistent across every SIM-enabled Android and iOS devices

SLC GUARDIAN ROOT LIVES HERE

LAYER 4 — Carrier Network

EAP-AKA · TS.43 · CAMARA APIs · Live network signals · Physical world anchor

Real-world carrier infrastructure — physically impossible to fake without being on the network

SLC operates in Layers 3–4 — independent of OEM, below the OS, rooted in the carrier network.

Built on Global Standards. Designed to Scale.



SLC is built on global telecom and identity standards—enabling a consistent, scalable identity layer across markets and devices.



Standards Based

- Built on GSMA, 3GPP, CAMARA frameworks
- Aligned with global telecom and identity standards



No OEM Dependency

- Works consistently across devices
- No reliance on manufacturer-specific integrations



Globally Deployable

- Leverages existing carrier infrastructure across 180+ countries
- Designed for scale across regions and markets



Real-Time Trust Signals

- Consistent identity layer across all users
- Works across devices and ecosystems



How SLC Guardian Works

Deterministic trust — hardware-anchored, carrier-verified, real-time

01



Hardware-Rooted Identity

ID as a cryptographic key bound into the SIM chip itself. Impossible to steal, copy, or spoof

- **Java Card applet** installed in eSIM Secure Execution Environment (SEE) on any android
- **ECDSA key pair** generated at onboarding — private key never leaves hardware
- **IMEI + IMSI** cryptographically bound to identity at enrollment

02



Tamper-Proof Auth Channel

Auth commands travel below the OS — invisible to malware

- **Class 2 SMS* channel** — carrier-routed, OS-blind, malware-resistant
- **Silent verification** (zero friction) or step-up with PIN/consent via native messaging
- **OS never sees the auth command** — cannot be intercepted at OS or app layer

03



Live Network Intelligence

Every auth verified against real-time carrier signals — not a stale database lookup

- **Real-time carrier signals** — not database queries or retrospective inference
- **SIM continuity** + cell tower presence confirms device is physically live
- **IMEI/IMSI binding integrity** validated at every single authentication event



SLC eSIM

SMS Types*:

Class 0 – Flash messages

Class 1 – Standard SMS

Class 2 – SIM-stored SMS

How Deterministic Trust Works

SLC's End-to-End Flow



Global Coverage

180+ countries

Telecom access to 600+ network operators

Identity access through IDEMIA across 2,400+ enterprises



SLC Platform

Provisions secure SIM or eSIM applets, routes transactions via API, and performs cryptographic attestation.



Enterprise Clients

Banks, fintechs, remittance providers, crypto platforms, and enterprises integrate SLC via API for high-risk transactions.



Activate eSIM

Users activate a secure SIM or eSIM applet via QR code or link to establish hardware-rooted identity.



Deterministic Signals

Device + Network Attestation including cell location, secure out-of-band messaging



Prime Data for Decision Engines

Enterprise adds SLC signals to their risk engine

SLC turns every device into a deterministic trust anchor for high-risk digital transactions.

Built on GSMA standards

Initial Use Cases



Wallet Login (Account Access)

- Deploy SLC identity layer across core wallet flows
- Integrate into authentication and transaction journeys
- Enable real-time proof of possession at scale

Immediate impact: Reduce ATO risk



High-Risk Transactions

- Withdrawals, transfers, large payments
- Step-up authentication where risk is highest

Immediate impact: Prevent high-value fraud events



Step-Up Authentication Replacement

- Replace SMS OTP in sensitive flows
- Reduce dependency on interceptable channels

Immediate impact: Lower cost + improved experience

Start with the highest-risk, highest-impact flows—then expand across the wallet.

Integration: Designed for Minimal Disruption



Built to Layer Into Existing Architecture

- SDK-based integration
- Works alongside current authentication methods
- No need to replace existing systems



Flexible Deployment Model

- Start with specific flows (login, step-up)
- Expand gradually across user journeys
- Controlled rollout by segment or geography

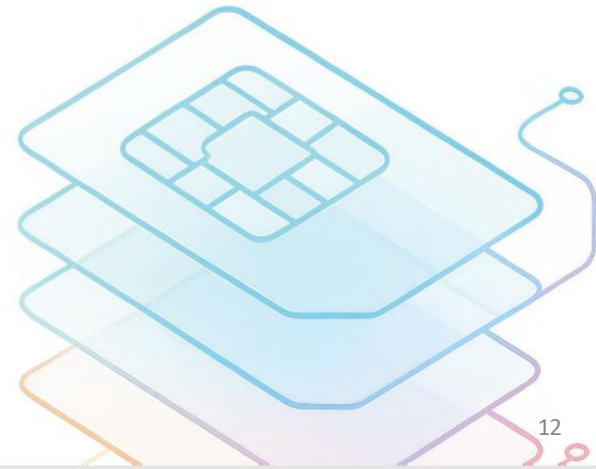


No Device or OEM Dependency

- Works across devices globally with SIM capability
- No reliance on manufacturer-specific hardware

SLC SDK integrates into existing stack for silent authentication

Supporting Materials



SLC is Powered by Global Telecom Infrastructure



SLC is deployed on top of global telecom infrastructure through Monogoto—enabling **real-time identity verification at global scale**.

Monogoto Platform



Global cellular connectivity platform (IoT + enterprise)



API-driven infrastructure for real-time integration



Built for scalable, programmable network access

Global Scale



180+ countries coverage
Global reach across major regions



Multi-carrier network access
Direct connectivity to leading MNOs worldwide



Millions of connected devices
Powering IoT and enterprise solutions at scale



Real-time network signaling and control
Reliable, secure, and low-latency delivery

Ecosystems & Marketplaces Powered by Monogoto Infrastructure



Through platform and ecosystem integrations



What This Enables



Real-time authentication
tied to live network presence



Global rollout
without building
telecom infrastructure



Consistent identity verification
across all users and regions



Built on existing global telecom scale—not new infrastructure.

Why now: Fraud and Regulation Are Forcing a New Identity Standard



Identity and fraud are now top enterprise priorities

-  **Account takeover attacks are surging**
Driven by phishing, SIM swaps, and device-based impersonation.
-  **Synthetic identity fraud is accelerating**
Now one of the fastest-growing fraud categories globally.
-  **Mobile and digital channels are the primary attack surface**
Where SMS, apps, and browsers are easiest to compromise.



Regulatory Inflection

-  **United States (NIST)**
Out-of-band SMS authentication is now restricted.
-  **European Union (PSD2 / SCA)**
SMS OTP is no longer sufficient alone.
-  **United Kingdom (NCSC)**
SMS ranked a weak authentication factor.
-  **UAE (Central Bank)**
SMS OTP replaced with biometric or app-based authentication.

SLC Patent Portfolio

SLC's intellectual property portfolio spans secure identity, authentication, and digital transaction technologies, providing a strong foundation for defensibility and differentiation



Hardware-Anchored Trust

SIM/eSIM and silicon-based methods for KYC/KYT and identity attestation.

- SLC US & PCT 18/610,571 — Secure-core silicon endpoints for KYC/KYT
- SLC US63/717,434 (Provisional) — RF triangulation with silicon endpoints & MEMS for identity/item validation



Financial Innovation

Stablecoin and fiat-currency stabilization mechanisms, supporting secure blockchain-based transactions.

- SLC US 18/610,571 (Provisional) — Stablecoin/fiat stabilization + secure blockchain smart contract execution



Next-Gen Authentication

Novel approaches using advanced materials (graphene) and sensory methods.

- SLC US & PCT 18/809,607 — Graphene-based human authentication via optical, ultrasound, and capacitance signals



Self-Sovereign Identity

Provisionals establishing frameworks for global, interoperable digital identity verification.

- SLC P006-US-P1 (Provisional) — Identity attestation & transactional communication in a self-sovereign digital identity framework



Secure Data Transmission

Patents covering encrypted, spatially-encoded, and radio-frequency communication techniques.

- SLC US 63/660,278 (Provisional) — Transmission of sensitive data through spatially-encoded sound



Standards & Compliance

SLC is built on established standards — not proprietary architecture

Standard	Status	Relevance to SLC
NIST SP 800-63	<i>AAL2 / AAL3</i>	Identity assurance level — hardware-bound authenticator characteristics
FIDO2 / WebAuthn	<i>Aligned</i>	Same cryptographic model — asymmetric key pair, challenge/response, private key in hardware
GSMA SGP.22	<i>Compliant</i>	Consumer eSIM standard — SLC provisioning operates within GSMA SGP.22 and SGP.32
3GPP TS 03.48/23.048	<i>Compliant</i>	OTA channel encryption and APDU command structure
CAMARA APIs	<i>Integrating</i>	CAMARA Number Verification, SIM Swap, Device Location
MITRE ATT&CK	<i>Mapped</i>	Four documented attack techniques addressed at hardware layer: T1451, T1630.003, T1539, T1638
ISO 27001:2022/NIST CSF 2.0	<i>Aligned</i>	Information Security Management System — organizational security program
ECDSA / ECDH	<i>Compliant</i>	Elliptic curve asymmetric keys stored on secure silicon
GDPR/CCPA	<i>Compliant</i>	Data privacy standard
ETSI/Global Platform	<i>Compliant</i>	Global telecom standards

Experienced Leadership Team

Deep Expertise Across Banking, Telecom, and Technology



CEO & Co-Founder
Travis M. McGregor

Founder and CEO of SLC Digital, leading the development of SIM-based digital identity and hardware-rooted authentication. A telecommunications and security veteran with 30+ years of experience, former CTIA Fraud Task Force member, and holder of foundational patents in wireless QoS, SIM-centric commerce, and biometric authentication. His work focuses on building globally scalable, phishing-proof trust infrastructure.



Chief Operating Officer
Veena Dandapani

Senior executive with 25+ years of experience across fintech, payments, and technology companies. Former Chief Client Services Officer at Pagaya and Global Head of Products at Western Union, scaling global digital platforms and leading enterprise-grade transformation. At SLC, she leads enterprise go-to-market, strategic partnerships, and operational scaling.



Chief Technology Officer
Sundaresh Sathasivan

Sundaresh Sathasivan is Chief Technology Officer at SLC Digital, leading engineering, cloud architecture, and AI strategy. Former Head of Enterprise Architecture and AI/ML at Western Union, specializing in secure, scalable systems for regulated environments. At SLC, he leads technology, security architecture, cryptographic trust systems, and telecom-native identity infrastructure.



Head of Product
Numan Maloney

Numan Maloney is Head of Product at SLC Digital, leading product strategy, platform design, and go-to-market use cases for its hardware-rooted identity infrastructure. With a background in business transformation and digital modernization consulting, he brings a systems-level perspective to scaling secure identity platforms globally. Formerly at Alvarez & Marsal, his experience spans financial services, government, and enterprise.

Board Of Directors



Co-Founder/Director
Travis M. McGregor

SLC Co-Founder

CEO of SLC, telecom pioneer with 30+ years founding companies, securing 75+ patents, and scaling mobile authentication platforms for global carriers and government.



Co-Founder/Director
Eric Maloney

SLC Co-Founder

Technology Strategist with 25+ years at Microsoft, JPMorgan Chase, Morgan Stanley, KPMG, and other financial leaders.



Director
Adrian E. Martinez

Ex. Chief Risk Officer, Citi

Senior financial executive with 30+ years across banking, payments, and fintech, including leadership at Citi and HSBC.



Director
Rob Furst

CEO Furst Capital

40+ years as investor, CEO, and advisor in venture capital, private equity, and commodities trading.



Director
Sam Graziano

Managing Director at Chatham Financial

Financial.Fintech operator and growth leader; former CEO of Foundation



Director
Paul Ostling

Ex. COO Ernst & Young.

CEO, COO, and board leader with global high-tech experience, corporate governance expertise, major transaction success, and Fordham Law background.



SLC SIM Root-of-Trust™

